



Overview of Basic Criteria and Models Applicable to Emergency Medical Service System Optimization

Marek Kvet¹

Received: 18 June 2022 / Accepted: 4 November 2022 / Published online: 30 November 2022
© The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd 2022

Abstract

This paper addresses the problem of searching for the optimal Emergency Medical Service stations deployment making use of the mathematical programming tools. Since the problem may be managed in many different ways, the attention is paid to the most common objectives, which are usually optimized. In addition, the associated modeling approaches connected with studied concepts are discussed. There are reported, analyzed and compared several mathematical models of the decision problems described by means of linear integer programming. To obtain the optimal solution of the problems, the integer programming solver XPRESS was used. This universal software tool enables solving various mathematical programming problems without the necessity of specific knowledge of Applied Informatics. The main benefit of this manuscript consists in a brief summary of possible approaches applicable to public service system designing. This research paper is dedicated to all operations researchers, mathematicians, programmers and to every public representative or practitioner, who is responsible for making strategic decision not only in health sector.

Keywords Location problems · Mathematical modelling · Design quality criteria comparison · EMS system · XPRESS

Introduction

Mathematical modelling, programming, optimization and the whole operations research field play a very important role in many areas of human life without us realizing it. Information technologies and various background algorithms accompany us almost every day. Results of particular research can be found in transport, industry, manufacturing systems, education, service systems management or in many other fields, in which big data need to be processed and stored [32–34]. In this extended conference paper, we concentrate on a special application field, which consists in urgent health-care management. Hereby, we deal with the problem of searching for the optimal network of Emergency Medical Service (EMS) stations, from which the paramedics

or doctors are sent to satisfy the randomly occurred demand. The satisfaction of the demand consists in providing urgent care and/or transporting the patient to the nearest medical facility for further treatment and cure. Simply said, we concentrate here of the discrete facility location problem in the EMS sector. The mentioned problem of obtaining the optimal service center deployment may take several forms [1, 20, 22]. Let us emphasize the most important and interesting modelling and solving concepts making use of different quality criteria of the system design.

The design of any service system including also the EMS stations network must usually adapt to several restrictions. Among the most known and the most common restrictions are limited human, material and technological resources or the financial budget. Thus, limited number of service centers often forms the system structure and the associated objective is to minimize some form of total disutility like social costs, which are proportional to the distances travelled by all users to the nearest source of provided service. The mentioned disutility can be expressed also by the time that elapses from the request to the arrival of rescuers at the scene of the event [7, 9, 23]. Obviously, the closer is the service center; the lower is the perceived disutility. Especially in the urgent health-care systems, it is necessary to satisfy the demand

This article is part of the topical collection “Advances on Operations Research and Enterprise Systems” guest edited by Marc Demange, Federico Liberatore and Greg H. Parlier.

✉ Marek Kvet
marek.kvet@fri.uniza.sk

¹ Faculty of Management Science and Informatics, University of Žilina, Univerzitná 8215/1, Žilina, Slovakia

immediately or within the shortest possible time, because the response time may play an essential role to avoid death or serious consequences to health [14, 36].

Mathematical model of the problem of finding the resulting service center deployment often takes the form of a weighted p -median problem, in which the number of system users to be served may reach several thousand [3]. Under the assumption that a facility (in our case EMS station) may be located in any network node corresponding to a dwelling place, the number of candidates can take the value of several thousand as well. Obviously, the mathematical model filled with real data represents a challenge for operations researchers and experts in algorithms to develop fast and effective solving techniques. There is no doubt that a concrete model form plays a decisive role in the speed of optimization method convergence. Another possibility for getting the results quickly enough consists in using some of advanced approaches, which will be discussed together with the particular model formulations.

Generally, whenever a service system, including also the EMS, is designed or optimized (here, the designing of a service system does not necessarily mean establishing a completely new system), then various quality criteria of the design may be applied. These criteria define the optimal solution and they can significantly influence the efficiency of service providing and other performance characteristics. Therefore, the main goal of this contribution is to compare selected criteria from different points of view. We study here their impact on the resulting system design. Thus, the comparison is the core extension and complement of the former study reported in [28].

As far as the system design quality criteria and modeling concepts resulting from them are concerned, available literature reports many linear programming models used to optimize EMS stations locations. From the point of concrete forms, there are plenty of them. Let us mention at least two basic concepts.

The first one consists in so-called *coverage* approach, the aim of which is to locate centers to cover all expected demands within given radius. The second concept follows from the idea of minimizing the average response time for all system users. The studies reported in [23, 24] were aimed at comparison of both mentioned concepts. Based on performed experiments and verification by a computer simulation it was found that the minimal average response time objective function produces better results than the maximal coverage approach. Therefore, we restrict ourselves only to such model objectives, which are based on the response time (may be proportional to distance) optimization. Let us look at some objectives in more details.

A common or “system” approach to EMS system design consists in the deployment of either exactly or at most given integer number p of service centers at some places from a

finite set of possible candidate locations so that the sum of disutility values perceived by all system users is minimal [3, 11, 26]. Optimization of the average response time may lead to the following inconvenience. Even if all system users share the costs for the system construction, not all users have the same access to the service. Especially, users from rear populated dwelling places do not consider this solution fair, because their contributions to the system costs are equal to the contributions of the others, but their distance from the nearest located service center is much higher in comparison to an average disutility in the entire system. To comply with the users’ call for equity in the service accessibility, various solidarity criteria have been developed [4, 6, 27] and some of them are reported and compared in this scientific paper. Presented models are based on commonly used location-allocation formulation. Of course, some possibilities for model simplifications are introduced.

For completeness, let us emphasize that this journal paper represents an extension and complementary study to the original contribution presented at ICORES 2022 and published in the conference proceedings [28]. While the original paper was aimed at a two-phase approach optimizing the network of Emergency Medical Service stations considering certain form of capacity restrictions, this new research paper focuses more on the objectives, which may be used in the mathematical models and may significantly affect the resulting optimal service center deployment. The extended version of the paper differs from the original mainly in that the original one did not contain any analysis of used objective function. Mentioned weakness can be solved by theoretical overview of basic quality criteria, which are compared in a computational study reported in this paper. Definition of various objectives may bring several new concepts of modelling, to which the existing methods cannot be easily adjusted. Therefore, we present here some challenges deserving future research and development of effective solving methods.

The structure of the remaining part of this research paper is organized according to the following description. The next chapter brings the basic definition of the weighted p -median problem with standard min-sum objective. There is also studied a possible modification of the objective function, which enables to minimize total disutility only of those users, whose discomfort exceeds given limit. In order to make the problem easily solvable by means of common optimization environment without the necessity of developing special software tool the reformulation of the problem into so-called radial form is introduced. Furthermore, a generalization of the approach for more centers providing service to the same client is explained. The third section is devoted to the description of the min-max approach, which follows the idea of minimizing the maximal discomfort that means the distance of the worst situated service recipients from the

center assigned to satisfy them. The original location-allocation model is compared to the radial one and the possibilities to speed-up the associated solving process by iterative solving of a simple covering model are discussed. In Section “Comprehensive Approaches and Future Challenges”, the strongest method from the point of fairness is studied. This approach is based on the Ogryczak’s idea of lexicographic optimization [35]. Furthermore, several advanced optimization approaches are suggested. Since they represent specific and nontrivial concepts, their solving deserves development of sophisticated algorithms. That is why these approaches are studied here only theoretically without computational experiments. The last two chapters are reserved for a computational study with the data of the self-governing region of Žilina, in which also our university is located, and to the concluding remarks. Some newly identified challenges for future research are also mentioned here.

Basic Min-Sum Approach and Its Modification

The EMS system design problem, in which the total users’ disutility is supposed to be minimal, can be formulated in the following way. The goal is to choose exactly p elements (p is a positive integer value) from given candidates set to achieve the smallest possible value of the optimization criterion. As far as a concrete form of the used criterion is concerned, it can be expressed as a sum of individual disutility values of each service recipient following from the nearest located station is minimal. This way, the common modelling approach considers so-called average service recipient.

Mathematical formulation of the above-mentioned problem requires the following notations. Let I be the set of candidates for locating facilities. Furthermore, let J represent the set of possible users’ locations (service recipients). Let us note that a specific network point represents each client location. Obviously, the sets I and J may be the same. The number of individual service recipients located at $j \in J$ will be denoted as b_j . This integer non-negative coefficient may have many other interpretations, i.e. the number of expected demands for service during some period. However, it can be understood as a weight of the location j . The disutility for a user located at j following from the possible center location i will be denoted as non-negative d_{ij} . The value of d_{ij} does not have to be necessarily integer, even if integer values have certain advantages, which will be explained later. The above-described problem is known as the weighted p -median problem, to which much attention is paid in the available literature mainly from the point of solving techniques suggested for large instances [5, 8, 28].

In the weighted p -median problem, the strategic decision about locating at most p facilities from the set I needs to be

made to optimize given criterion. Usually, the most common objective function expresses the sum of user disutility contributions following from the distance to the nearest located facility or from the time, the nearest center can be achieved in.

Mentioned strategic decision on service center location can be made using a bivalent variable $y_i \in \{0, 1\}$, which equals one, if a facility is to be located at $i \in I$, and it takes the value of zero otherwise. To complete the mathematical model, allocation variables $z_{ij} \in \{0, 1\}$ are introduced for each $i \in I$ and $j \in J$ to assign a service recipient located at j to a facility location i by setting the value of one.

To meet the problem requirements, the variables y_i and z_{ij} must satisfy the following constraints. Then, expressions (1)–(6) represent the location-allocation model of the problem.

$$\text{Minimize} \quad \sum_{i \in I} \sum_{j \in J} b_j d_{ij} z_{ij}, \quad (1)$$

$$\text{Subject to :} \quad \sum_{i \in I} z_{ij} = 1 \quad \text{for } j \in J, \quad (2)$$

$$\sum_{i \in I} y_i = p, \quad (3)$$

$$z_{ij} \leq y_i \quad \text{for } i \in I, j \in J, \quad (4)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I, \quad (5)$$

$$z_{ij} \in \{0, 1\} \quad \text{for } i \in I, j \in J. \quad (6)$$

Even if the location-allocation model of the weighted p -median problem is generally known, let us remind the meaning of above-mentioned expressions. The minimized objective (1) represents the sum of distances between service recipients and their nearest service providers. The allocation constraints (2) ensure that each service recipient is assigned to a possible service center. Obviously, the assignment of service recipients can be done only to located centers. This fact is guaranteed by the link-up constraints (4). The maximal possible number of located facilities is kept by (3). The obligatory constraints (5) and (6) are used to guard correct values of decision variables.

Since the weighted p -median problem and its model belong to the basics of integer programming with a wide range of practical applications, let us look at the model (1)–(6) in more details and let analyze its computational complexity. Consider a small example. If the cardinalities of the input sets I and J are 100, then it is necessary to introduce exactly 10.000 allocation variables, but only 100 of them take the value of one. The number of allocation variables z_{ij} corresponds to the

size of the distance matrix $\{d_{ij}\}$. The bigger is the matrix, the larger is the number of allocation variables to be introduced and the more difficult the problem gets. Furthermore, if we focus on the allocation constraints (2), then we can see that there is only one allocation variable for each $j \in J$ that equals one. It means that each service recipient is assigned only to one service center. From the viewpoint of objective function (1), it is not so important to know, which located service center is the nearest one for the user located at j , because the most important information is the distance. Based on these facts, the operations researchers, mathematicians and programmers have made a great effort to develop such model simplification to achieve better solvability of large instances. The solution of mentioned weakness of the location-allocation model (1)–(6) consists in so-called radial formulation, which was originally introduced in [11, 15, 16] and improved in [25, 26].

The core concept of the radial approach follows the idea that the direct assignment of service recipients to the located EMS stations by the allocation variables z_{ij} is not necessary and the distance from the customer located at j to the nearest center can be expressed making use of the following notations. Let the symbol m denote the highest value in the matrix $\{d_{ij}\}$, i.e. $m = \max\{d_{ij} : i \in I, j \in J\}$. Let us assume that the distances in the matrix are integer. Otherwise, the original radial model could be easily adjusted. Generalization of the approach for real values will be reported after the basic radial model. For each service recipients' location $j \in J$ and for each $v = 0, 1, \dots, m-1$ we introduce a binary variable $x_{jv} \in \{0, 1\}$. This auxiliary variable equals one, if the distance d_{j*} from the service recipient located at $j \in J$ to the nearest center is greater than v and it takes the value of zero otherwise. Then, the distance d_{j*} for each $j \in J$ can be rewritten to the form of (7).

$$d_{j*} = \sum_{v=0}^{m-1} x_{jv}. \quad (7)$$

In addition, a matrix $\{a_{ij}^v\}$ has to be evaluated by (8).

$$a_{ij}^v = \begin{cases} 1 & \text{if } d_{ij} \leq v \\ 0 & \text{otherwise} \end{cases} \quad \text{for } i \in I, j \in J, v = 0, 1, \dots, m-1. \quad (8)$$

Having introduced all necessary inputs, the radial model of the problem can take the form of (9)–(13).

$$\text{Minimize} \quad \sum_{j \in J} b_j \sum_{v=0}^{m-1} x_{jv}, \quad (9)$$

$$\text{Subject to :} \quad x_{jv} + \sum_{i \in I} a_{ij}^v y_i \geq 1 \quad (10)$$

$$\text{for } j \in J, v = 0, 1, \dots, m-1,$$

$$\sum_{i \in I} y_i = p, \quad (11)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I, \quad (12)$$

$$x_{jv} \in \{0, 1\} \quad \text{for } j \in J, v = 0, 1, \dots, m-1. \quad (13)$$

To explain the model by a brief summary, the minimized objective function (9) expresses the sum of distances from all users to their nearest located station. The relation between variables x_{jv} and y_i is controlled by the link-up constraints (10). If there is at least one center located in radius v from the location j , the variable x_{jv} can take the value of zero. The constraint (11) keeps holding the rule of not exceeding the highest number of located facilities. The last two sets of mandatory restrictions (12) and (13) check the domain of the variables y_i and x_{jv} . Presented radial formulation enables us to solve much larger instances than described only by the former location-allocation approach [11, 15, 26].

If the distances were not integer or one would be satisfied with a lower or upper approximation of distances, the radial formulation can be generalized in the following way. To obtain upper or lower approximations of the objective function (1), the range $[d_0, d_m]$ of all distance values $d_0 < d_1 < \dots < d_m$ from the matrix $\{d_{ij}\}$, where $d_m = \max\{d_{ij} : i \in I, j \in J\}$, is partitioned into $v+1$ zones as depicted in Fig. 1. The zones are clearly determined by a finite ascending sequence of values called dividing points. These bordering dividing points $D_1, D_2, \dots, D_v$ are chosen from the sequence of distances $d_0 < d_1 < \dots < d_m$, in which $0 = d_0 = D_0 < D_1$ and also $D_v < D_{v+1} = d_m$. The radial zones can be identified so that the zone s corresponds to the interval $(D_s, D_{s+1}]$. The length of the s th zone is denoted by e_s for $s = 0, \dots, v$. In addition, auxiliary binary variables x_{js} need to be introduced for each s . The variable x_{js} takes the value of one, if the distance of the service recipient located at $j \in J$ from the nearest located service provider is greater than D_s and it equals zero otherwise. Then the expression $e_0 x_{j0} + e_1 x_{j1} + \dots + e_v x_{jv}$ constitutes an upper approximation of the distance d_{j*} from the location j to the nearest located service center.

If the distance d_{j*} belongs to the interval $(D_s, D_{s+1}]$, then the value of D_{s+1} is an upper estimation of d_{j*} with

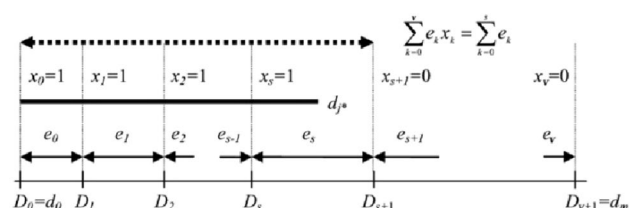


Fig. 1 Approximation of distances by dividing points [26]

the maximal possible deviation e_s . Obviously, the covering coefficients a_{ij}^s for each triple $[i, j, s]$, where $i \in I, j \in J$ and $s \in \{0, \dots, v\}$ need to be redefined. The constant a_{ij}^s is equal to one, if the distance d_{ij} between the user location j and the possible EMS station location i is less than or equal to D_s , otherwise $a_{ij}^s = 0$. Based on these preliminaries, the model (9)–(13) can be rewritten to the form of (14)–(18) as published in [16]. By suitable selection of dividing points (number of them and their values), we can affect the model size and accuracy of the result as well.

$$\text{Minimize} \quad \sum_{j \in J} b_j \sum_{s=0}^v e_s x_{js}, \quad (14)$$

$$\text{Subject to:} \quad x_{js} + \sum_{i \in I} a_{ij}^s y_i \geq 1 \quad (15)$$

for $j \in J, s = 0, 1, \dots, v$,

$$\sum_{i \in I} y_i = p, \quad (16)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I, \quad (17)$$

$$x_{js} \in \{0, 1\} \quad \text{for } j \in J, s = 0, 1, \dots, v. \quad (18)$$

The radial approach can be used also in such cases, if the service could be provided not only from the nearest located EMS station, but also from the nearest available one (does not have to be the nearest located). In such case, we talk about the concept of so-called generalized disutility [21, 25].

The main disadvantage of mathematical models for the design of large EMS systems lies in their inability to master the complexity of the modeled systems. Deterministic models do not reflect the actual way of handling the requirements of system users well, because the weighted p -median model assumes that the request is always served from the center that was assigned to it by solving the problem.

However, this assumption of static assignment is not met in operational management of real service systems, especially in health sector. It must be realized that the demands for service emerge randomly and they can usually not be expected. It follows from the above that at the time of the request, the nearest service center may not be available due to the provision of a service to another patient. In such a case, the demand is assigned to another service center that is free and near enough. Generally, it is assumed that r nearest located centers can participate in the provision of the service for patients. In the criterion description, q_k stands for probability of the case that the

k th nearest center is the closest one, which is accessible. This concept makes the model much more realistic. The main idea can be easily understood from Fig. 2.

Implementing the concept of general disutility into the model involves several steps. In the case of the radial formulation of the weighted p -median problem described above, it is still necessary to adjust the structural and obligatory constraints in addition to the objective function. The resulting model will have the following form.

$$\text{Minimize} \quad \sum_{j \in J} b_j \sum_{v=0}^{m-1} \sum_{k=1}^r q_k x_{jvk}, \quad (19)$$

$$\text{Subject to:} \quad \sum_{k=1}^r x_{jvk} + \sum_{i \in I} a_{ij}^v y_i \geq r \quad (20)$$

for $j \in J, v = 0, 1, \dots, m-1$,

$$\sum_{i \in I} y_i = p, \quad (21)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I, \quad (22)$$

$$x_{jvk} \in \{0, 1\} \quad \text{for } j \in J, v = 0, \dots, m-1, k = 1, \dots, r. \quad (23)$$

The objective function (19) expresses the expected volume of generalized disutility. The constraints (20) ensure that the sum of variables x_{jvk} over k expresses the number of the service centers in the radius v from the user location j , which remains to the number r . The constraints (21), (22) and (23) keep the same meaning as in the above mathematical models. The computational study and other characteristics of generalized disutility can be found in [21, 25]. Incorporating this concept into the mathematical model of

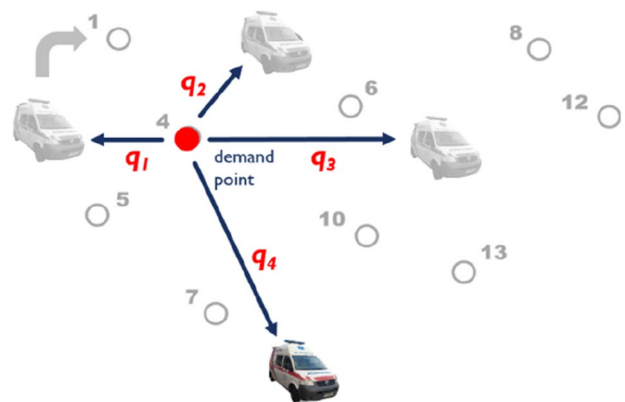


Fig. 2 Concept of generalized disutility, where the distances to the service centers are weighted by probability coefficients q_k . The shortest distance is weighted by the highest coefficient

the weighted p -median problem significantly increases the possibilities of its practical usage.

Min–Max Approach

The min–max approach solves those system users that are too far from any located service center and the associated solving process tries to choose the locations of service centers in such a way that the worst distance is minimized. Obviously, the model may use the same notations as introduced in Section “Basic Min–Sum Approach and its Modification”. The variables y_i and z_{ij} keep the same meaning as above. To solve the problem, at most p locations from the set I must be chosen so that the maximal disutility perceived by the worst situated user is minimal. In order to be able to formulate a model with an objective function of the *minMax* type, it is necessary to introduce a non-negative variable h , which will represent the minimized upper bound of distances between patients and their nearest service center. Particular location-allocation model of this problem can be formulated in the form of (24)–(31).

$$\text{Minimize } h, \quad (24)$$

$$\text{Subject to : } \sum_{i \in I} z_{ij} = 1 \quad \text{for } j \in J, \quad (25)$$

$$\sum_{i \in I} y_i = p, \quad (26)$$

$$z_{ij} \leq y_i \quad \text{for } i \in I, j \in J, \quad (27)$$

$$\sum_{i \in I} d_{ij} z_{ij} \leq h \quad \text{for } j \in J, \quad (28)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I, \quad (29)$$

$$z_{ij} \in \{0, 1\} \quad \text{for } i \in I, j \in J, \quad (30)$$

$$h \geq 0. \quad (31)$$

Models (24)–(31) does not require special explanation, because it is based on models (1)–(6). The objective function (24) represented by single variable h gives the upper bound of all perceived disutility values. The constraints (28) ensure that each perceived disutility is less than or equal to the upper bound h .

From the practical point of view, there are two main issues to be discussed. The first problem follows from the necessity of allocation variables z_{ij} as in models (1)–(6). Thus, solving of the real-world instances demands usually

unacceptably much memory. Similarly to the location-allocation model, the min–max models (24)–(31) can be easily reformulated into the form of (32)–(38).

$$\text{Minimize } h, \quad (32)$$

$$\text{Subject to : } x_{jv} + \sum_{i \in I} a_{ij}^v y_i \geq 1 \quad (33)$$

$$\text{for } j \in J, v = 0, \dots, m-1,$$

$$\sum_{i \in I} y_i = p, \quad (34)$$

$$\sum_{v=0}^{m-1} x_{jv} \leq h \quad \text{for } j \in J, \quad (35)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I, \quad (36)$$

$$x_{jv} \geq 0 \quad \text{for } j \in J, v = 0, \dots, m-1, \quad (37)$$

$$h \geq 0. \quad (38)$$

The objective function (32) expresses the upper bound of all patients' distances from their nearest stations and the series of constraints (35) defines the relation between mentioned distances and the corresponding value of h .

Even if the radial formulation is generally better and avoids introducing allocation variables, in the case of the min–max model its advantageous properties were not fully manifested. Despite a significant reduction in computational time can be achieved compared to the location-allocation approach, the problems with high computational time requirements persist. Slow convergence of the branch and bounds method, which is the basic solving algorithm for integer linear programming problems, could be explained by binding structural constraints (35). If we look in detail at the structure of the model, we find that the radial formulation of the original model allows a relatively simple transformation of the problem to a simpler form without the necessity of introducing the upper bound h and the constraints (35). The essence of the transformation lies in the usage of advantageous properties of covering problems and it is based on the following assumption:

The values of the variables x_{jv} for $j \in J$ and $v = 0, 1, \dots, m-1$ form a vector with a specific structure. If we sort the variables x_{jv} according to the index v from 0, then the corresponding vector will have the form $(1, 1, \dots, 1, 0, 0, \dots, 0)$. Thus, at the beginning, the variables will be equal to one and at some point, they will change their values, being zero from this index to the end [11, 15, 26]. This fact is then key in the transformation of the min–max model into a covering one.

Then, instead of solving the model (32)–(38), we achieve the minimization of the largest distance between patients and their nearest EMS stations as follows: From the sequence of distances $d_0, d_1, \dots, d_m$ we choose the “middle” dividing point D_s . For this dividing point, we will solve a simple covering problem (39)–(43), on the basis of which we will find out whether there is such a feasible solution of the problem that all distances between patients and their service centers are less than or equal to the value of D_s . In such a case, all variables x_j would take the value of zero. Otherwise, there would be at least one non-zero variable x_j . We determine the covering coefficients a_{ij} for the given D_s according to the following rule. If $d_{ij} \leq D_s$, then $a_{ij} = 1$, else $a_{ij} = 0$. Then the mathematical model for the dividing point D_s can be expressed by (39)–(43).

$$\text{Minimize} \quad \sum_{j \in J} x_j, \quad (39)$$

$$\text{Subject to :} \quad x_j + \sum_{i \in I} a_{ij} y_i \geq 1 \quad \text{for } j \in J, \quad (40)$$

$$\sum_{i \in I} y_i = p, \quad (41)$$

$$y_i \in \{0, 1\} \quad \text{for } i \in I, \quad (42)$$

$$x_j \geq 0 \quad \text{for } j \in J. \quad (43)$$

If by solving models (39)–(43) we find that all variables x_j take the zero value by solving models (39)–(43), then we repeat the computation for a lower value of D_s , otherwise we choose a higher dividing point D_s . This way—by combining coverage problems (39)–(43) and the bisection method applied to the whole sequence of values $d_0, d_1, \dots, d_m$ —we can find the minimal possible distance of the worst patients from their nearest EMS stations. Solving the min–max problem using a repeated computation of a simple covering problem is much faster than solving the original min–max model even if it were formulated in the radial form.

Comprehensive Approaches and Future Challenges

This section is devoted to the theoretical explanation of some additional advanced schemes and future challenges.

The first concept combines system criterion (optimization of average response time or distance) with the request of fairness. It means, that we solve here a simple weighted p -median problem (9)–(13), to which we add simple constraints (44). This constraint ensures that each user will have

a center in the radius of at most DmM , where DmM denotes the result of min–max model.

$$\sum_{i \in I} a_{ij}^{\max} y_i \geq 1 \quad \text{for } j \in J. \quad (44)$$

The coefficients $a_{ij}^{\max}$ are determined by formula (45).

$$a_{ij}^{\max} = \begin{cases} 1 & \text{if } d_{ij} \leq DmM \\ 0 & \text{otherwise} \end{cases} \quad \text{for } i \in I, j \in J. \quad (45)$$

The composed approach was studied in [29]. The obtained results confirmed the expectation that by applying a two-phase method, better distribution of EMS stations from the patient’s point of view can be obtained in comparison to a simple min–max approach, which focuses on minimizing only the highest distance of the worst situated users from the nearest centers.

The second approach, which can be applied when an EMS system is to be designed or optimized, follows from a simple analysis of possible criteria. If we concentrated on all presented models, they follow from the weighted p -median problem. Any of mentioned models can be expressed in a simplified general form of (46), in which $f(\mathbf{y})$ represents the optimized objective function and $\mathbf{y}$ is the vector of location variables y_i for $i \in I$.

$$\min \left\{ f(\mathbf{y}) : \sum_{i \in I} y_i = p \right\}. \quad (46)$$

The biggest disadvantage of all the approaches consists in the fact that they are able to optimize only one objective. When an Emergency Medical Service system is to be optimized, there are usually plenty of people involved in the decision-making process. They form interest groups with different ideas about how the system should work effectively. This fact can significantly complicate the decision-making process. Consider the following example: if we optimized the average availability of a service for users by minimizing the shortest time in which the service could be provided (so-called *system* criterion), there would certainly be a small group of people too far away from the nearest source of provided service. Such a system design would be considered unfair. On the other hand, if we minimized only the number of those who are disadvantaged by their location (so-called *fair* criterion), we would worsen the system criterion. Thus, these two mentioned simple objectives are contradictory [19].

Therefore, a big challenge for the operations researchers, mathematicians and experts in Applied Informatics consists in developing modeling and solving techniques, which would enable us to obtain suitable results of those problems, in which at least two different criteria need to be considered. Since the criteria may be in a conflict, one system design

does not hold as proper output of optimization. As the set of all feasible solution is usually extremely large, the attention of scientists is paid to so-called Pareto front of solutions, which represent a small selection of possible results [2, 19]. The recent attention of the operations researchers is paid to the development of several advanced heuristic approaches [10, 12] for Pareto front approximation, because obtaining the complete Pareto front is very time demanding [13, 17].

Another modelling strategy needed to be mentioned here follows from the idea, that the Emergency Medical System or any other public service system may be optimized to increase the level of system performance efficiency of to improve the accessibility of provided service, but such systems already exist and they are not to be newly established. From this point of view, it is necessary to take into account current service center deployment. Therefore, proposed changes following from the optimization process must be limited. Otherwise, they will not be accepted by public. People are generally very sensitive to changes in service systems, mainly in health sector. Mentioned modelling strategy is usually referenced as so-called system reengineering [18, 30, 31].

Computational Experiments

The computational study is concentrated on the practical usage of presented models to get real EMS system design. The whole study was performed using FICO Xpress 7.7 (64-bit, release 2014). The associated models were run on a common PC equipped with the Intel® Core™ i7 2630QM 2.0 GHz processor and 8 GB RAM. The approaches were tested on the benchmarks obtained from the road network of Slovakia, on which the emergency Medical Service system is operated. Even if we used the datasets of all regions, presented tables contain only the results for the self-governing region of Žilina. Other results proved to be very similar. Therefore, we decided not to report large tables full of numbers.

The instance of Žilina self-governing region contains 315 network nodes, in which an EMS station can be located and those network nodes represent also the locations of served population. The self-governing region of Žilina is situated in the north-western part of Slovak Republic. The region comprises five subregions (Horné Považie, Kysuce, Liptov, Orava and Turiec) that are divided into 11 districts. Covering an area of 6809 km² (14% of the territory of Slovakia), the region has a total population of 690 778 and a population density of 101.4 inhabitants/km². The location of Žilina region within the country of Slovakia is depicted in Fig. 3.

The coefficients b_j used in the models represent the number of inhabitants and these values were rounded to hundreds.



Fig. 3 Self-governing region of Žilina

The reported approaches with different quality criteria were compared from various points of view. The standard system approach described by the model (1)–(6) was taken as the referential, to which the other models were compared. For each resulting EMS system design obtained by particular method, several characteristics were computed. First, we wanted to evaluate the loss of the system criterion value caused by adopting the approach to more affected users. Therefore, we evaluated the system objective function $F(\mathbf{y})$ as a sum of disutility values perceived by all system users. Its value can be expressed by (47) for the arbitrary solution $\mathbf{y} = [y_1, y_2, \dots, y_{|I|}]$, where y_i takes the value of one, if a service center is located at the location i and it takes the value of zero otherwise.

$$F(\mathbf{y}) = \sum_{j \in J} b_j \min \{d_{ij} : i \in I, y_i = 1\}. \quad (47)$$

If $\mathbf{y}^s$ denotes the vector of location variables y_i obtained by solving the model (1)–(6), then we can evaluate the loss of objective function value in a general fair approach by so-called *price of fairness*. To formulate the *price of fairness* POF, it is necessary to denote the vector of location variables y_i in the fair model by $\mathbf{y}^f$. Then the relation between the system and fair optimum called *price of fairness* denoted by POF can be expressed in percentage of the system objective function.

$$\text{POF} = 100 \frac{F(\mathbf{y}^f) - F(\mathbf{y}^s)}{F(\mathbf{y}^s)}. \quad (48)$$

Another characteristic of any public service system design can be defined by the maximal relevant distance from the worst situated service recipients. This value is denoted as DmM and mathematically it can be formulated by (49).

$$DmM = \max \{d_{ij} : i \in I, j \in J, y_i = 1\}. \quad (49)$$

In addition, we can compare the resulting system structures from the viewpoint of different solutions by *Hamming distance* of location vectors $\mathbf{y}^s$ and $\mathbf{y}^f$ obtained for system and

fair design. The *Hamming distance* $H(\mathbf{y}^s, \mathbf{y}^f)$ represents the “Manhattan distance” between two tops of unit hypercube and it can be computed according to (50). In the tables with our achieved results, this value is denoted as *HD*.

$$H(\mathbf{y}^f, \mathbf{y}^s) = \sum_{i=1}^m |y_i^f - y_i^s| \quad (50)$$

The first set of experiments was performed to study the efficiency of the “system” approach, which solves the location-allocation model (1)–(6) with its possible modification, in which we used coefficients c_{ij} instead of the former d_{ij} . The coefficients c_{ij} were set in the following way. If $d_{ij} \leq 15$, then $c_{ij} = 0$, otherwise $c_{ij} = d_{ij}$. The achieved results are reported in Tables 1 and 2. The computational time in seconds is given in the columns denoted as *Time*.

The reported results are quite interesting. If we consider the accuracy of the results as concerns the system objective function value $F(\mathbf{y})$ computed according to (47), then we can see big differences in the designs. Especially, when the value of parameter p , which gives the number of located facilities, is high, then the modified model brings a solution, that differs from the “system” result given by the model (1)–(6) a lot. This fact can be explained very easily. The objective

function with coefficients c_{ij} minimizes the disutility only of those users, whose disutility exceeds given limit of 15 units (minutes or kilometers). Then, the associated solving algorithm tries to locate the service centers in such a way, that all users have some service center near enough (particular disutility is lower than or equal to 15 and the coefficient c_{ij} is equal to 0). In such a case, the solver does not look at the sum of disutility values to be minimal. Thus, the difference in service center locations is so huge and the value of *POF* is very high, too. On the other hand, if the value of parameter p is lower, than the studied min-sum approaches bring very similar results, what is confirmed by lower values of *POF* and *HD*.

The next portion of experiments was realized to verify the suitability of the min-max approach (24)–(31). The achieved results are reported in Table 3.

Conclusions

The objective of this scientific paper was to bring a brief overview of various possible objective functions and associated modeling approaches, which can be used, when an EMS or any similar system needs to be designed, optimized or newly established. Since basic models are often incorporated into more complex methods, it is important to know, what impact on the resulting system they may bring. There are reported, analyzed and compared several models of the decision-making problems described by means of linear programming. We did not report only the basic models themselves, but we have discussed possibilities of their improvement or reformulation to achieve better solvability. Since the practical problems bring several new requirements, we paid attention also to new challenges, which need to be solved in near future. We hope that we have filled an empty space in short objective analyses and extended the available methods.

One of the biggest disadvantages of almost all modeling and solving approaches reported in this paper consists in the fact that only one objective function is optimized at the same

Table 1 Results of the min-sum approach—model (1)–(6)

$ I $	p	Time	$F(\mathbf{y})$	DmM
315	158	5.80	2444	13
315	105	5.85	5594	17
315	79	6.75	8430	20
315	63	6.35	11,125	20
315	32	5.80	20,995	26
315	21	5.77	28,548	26
315	16	5.98	34,405	33
315	11	6.01	47,092	33
315	8	5.94	58,665	42
315	7	5.99	63,865	42

Table 2 Results of the modified min-sum approach

$ I $	p	Time	$F(\mathbf{y})$	POF	HD	DmM
315	158	5,23	14,528	494,44	154	12
315	105	5,13	19,155	242,42	130	13
315	79	7,07	26,483	214,15	112	15
315	63	9,98	25,978	133,51	100	15
315	32	17,71	34,366	63,69	46	15
315	21	9,98	43,023	50,70	30	26
315	16	10,50	45,167	31,28	18	28
315	11	7,39	63,881	35,65	16	36
315	8	6,97	62,895	7,21	8	42
315	7	7,36	72,698	13,83	10	42

Table 3 Results of the min–max approach

$ I $	p	Time	$F(y)$	POF	HD	DmM
315	158	23.10	9199	276.39	118	4
315	105	15.46	13,830	147.23	115	6
315	79	33.71	21,878	159.53	102	7
315	63	33.65	24,009	115.81	78	8
315	32	60.86	32,680	55.66	44	14
315	21	48.49	49,993	75.12	34	16
315	16	83.27	47,602	38.36	20	20
315	11	79.75	77,664	64.92	20	25
315	8	133.22	99,000	68.75	14	31
315	7	126.28	112,530	76.20	12	34

time. Emergency Medical System and many other services are much more complex and such a simplification on one objective is too strict. Therefore, a future research challenge for the operations researchers, mathematicians and experts in Applied Informatics consists in developing such modeling and solving techniques, which would enable us to obtain suitable results of problems, in which at least two different criteria need to be considered simultaneously. Since the criteria may be in a conflict, so-called Pareto front of solutions could be given as the output. One of the nearest research goals could be aimed at the development of exact and heuristic approaches for bi- or even multi-criteria optimization.

Another big research topic could be aimed and creating a complex optimization and simulation software tool, which could verify the results of optimization by a computer simulation, which could help us to find some weaknesses of mathematical models to generate a suitable and complex optimization method.

Acknowledgements This work was supported by the research grants VEGA 1/0216/21 “Design of emergency systems with conflicting criteria using artificial intelligence tools” and VEGA 1/0654/22 “Cost-effective design of combined charging infrastructure and efficient operation of electric vehicles in public transport in sustainable cities and regions”. This work was supported also by the Slovak Research and Development Agency under the Contract no. APVV-19-0441.

Data availability The data, which were used in presented computational study are available online at <http://frdsa.fri.uniza.sk/~betka/indexPovodny.html>.

Declarations

Conflict of interest The author of the manuscript declares that not to have any conflicts of interest to report. Presented results do not contain studies with human or animal subjects.

References

- Ahmadi-Javid A, Seyedi P, et al. A survey of healthcare facility location. *Comput Oper Res*. 2017;79:223–63.
- Arroyo JEC, dos Santos PM, Soares MS, Santos AG. A multi-objective genetic algorithm with path relinking for the p-median problem. In: *Proceedings of the 12th Ibero-American Conference on Advances in Artificial Intelligence*, 2010; p. 70–9.
- Avella P, Sassano A, Vassil'ev I. Computational study of large scale p-median problems. *Math Program*. 2007;109:89–114.
- Bertsimas D, Farias VF, Trichakis N. The price of fairness. *Oper Res*. 2011;59(2011):17–31.
- Brotcorne L, Laporte G, Semet F. Ambulance location and relocation models. *Eur J Oper Res*. 2003;147:451–63.
- Buzna E, Koháni M, Janáček J. Proportionally fairer public service systems design. *Commun Sci Lett Univ Žilina*. 2013;15(1):14–8.
- Cornuéjols G, Nemhauser GL, Wolsey LA. A canonical representation of simple plant location problems and its applications. *SIAM J Algebraic Discrete Methods*. 1980;1(3):261–72.
- Current J, Daskin M, Schilling D, et al. Discrete network location models. In: Drezner Z, et al., editors. *Facility location: applications and theory*. Springer; 2002. p. 81–118.
- Doerner KF, Gutjahr WJ, Hartl RF, Karall M, Reimann M. Heuristic Solution of an Extended Double-Coverage Ambulance Location Problem for Austria. *CEJOR*. 2005;13(4):325–40.
- Drezner T, Drezner Z. The gravity p-median model. *Eur J Oper Res*. 2007;179:1239–51.
- García S, Labbé M, Marín A. Solving large p-median problems with a radius formulation. *INFORMS J Comput*. 2011;23(4):546–56.
- Gopal G. Hybridization in genetic algorithms. *Int J Adv Res Comput Sci Softw Eng*. 2013;3:403–9.
- Grygar D, Fabricius R. An efficient adjustment of genetic algorithm for Pareto front determination. *Trans Res Proc*. 2019;40:1335–42.
- Ingolfsson A, Budge S, Erkut E. Optimal ambulance location with random delays and travel times. *Health Care Manag Sci*. 2008;11(3):262–74.
- Janáček J. Approximate covering models of location problems. In: *Lecture Notes in Management Science. Proceedings of the 1st International Conference on Applied Operational Research ICAOR '08*, Vol. 1, Sept. 2008, Yerevan, Armenia, p. 53–61.
- Janáček J. Lower bound on large p-median problem using the sequential zone adjustment method. In: *Mathematical methods in economics 2011, Proceedings of the 29th international conference: September 6–9, 2011, Janská Dolina, Slovakia. Part I and II*. Prague: Professional Publishing; 2011. ISBN 978-80-7431-059-1. p. 317–20.
- Janáček J, Fabricius R. Public service system design with conflicting criteria. *IEEE Access Pract Innov Open Sol*. 2021;9:130665–79 (ISSN 2169-3536).

18. Janáček J, Kvet M. Reengineering of the emergency service system from the point of service provider. In: ICORES 2018: 7th International Conference on operations research and enterprise systems, 2018; p. 224–31.
19. Janáček J, Kvet M. Emergency medical system under conflicting criteria. In: SOR 2021 Proceedings, 2021; p. 629–35.
20. Janáček J, Linda B, Ritschelová I. Optimization of municipalities with extended competence selection. *Prager Econ Papers-Qu J Econ Theory Policy*. 2010;19(1):21–34.
21. Jankovič P. Calculating reduction coefficients for optimization of emergency service system using microscopic simulation model. In: 17th International Symposium on Computational Intelligence and Informatics, 2016; p. 163–7.
22. Jánošíková L. Emergency medical service planning. *Commun Sci Lett Univ Žilina*. 2007;9(2):64–8.
23. Jánošíková L, Kvet M, Jankovič P, Gábrišová L. An optimization and simulation approach to emergency stations relocation. *Cent Euro J Oper Res*. 2019;27(3):737–58.
24. Jánošíková L, Jankovič P, Kvet M, Zajacová F. Coverage versus response time objectives in ambulance location. *Int J Health Geogr*. 2021;20:1–16.
25. Kvet M. Computational Study of radial approach to public service system design with generalized utility. In: Proceedings of International Conference DT 2014, Žilina, Slovakia, 2014; p. 198–208.
26. Kvet, M.: Advanced radial approach to resource location problems. In: Developments and advances in intelligent systems and applications. Cham: Springer International Publishing, 2018, Studies in computational intelligence, 718, 2018; p. 29–48.
27. Kvet M. Impact of fairness constraints on average service accessibility in emergency medical system. In: Information and Digital Technologies 2021, 2021; p. 11–8.
28. Kvet, M.: Optimization of emergency medical service with fixed centers. In: Proceedings of the 11th International Conference on Operations Research and Enterprise Systems (ICORES), 2022; p. 214–21.
29. Kvet M, Janáček J. Composed min-max and min-sum radial approach to the emergency system design. In: Operations Research Proceedings 2015. Springer; 2016. p. 41–7.
30. Kvet M, Janáček J. Reengineering of the Emergency Service System under Generalized Disutility. In ICORES 2018: 7th International Conference on Operations Research and Enterprise Systems, 2018; p. 85–93.
31. Kvet M, Janáček J, Kvet M. Computational study of emergency service system reengineering under generalized disutility. In: Operations Research and Enterprise Systems: 7th International Conference, ICORES 2018 Funchal, Madeira, Portugal, January 24–26, 2018 Revised Selected Papers: Springer; 2019. p. 198–219.
32. Kvet M, Matiaško K. Magnetic resonance imaging results processing: brain tumour market value processing. In: Digital technologies: proceedings of the ninth international conference: 29–31 May 2013, Žilina, Slovakia, Žilina: University of Žilina, 2013, ISBN 978-80-554-0682-4. p. 149–59.
33. Kvet M, Matiaško K. Epsilon temporal data in MRI results processing. In: Digital technologies: the 10th international conference: 9–11 July 2014 Žilina, Slovakia, 2014, ISBN 978-1-4799-3301-3. p. 209–17.
34. Marianov V, Serra D, et al. Location problems in the public sector. In: Drezner Z, et al., editors. Facility location applications and theory. Berlin: Springer; 2002. p. 119–50.
35. Ogryczak W, Sliwinski T. On direct methods for lexicographic min-max optimization. In: Gavrilova M, et al. editors. ICCSA 2006, LNCS 3982. Berlin Heidelberg: Springer-Verlag; 2006. p. 802–11.
36. Pardede AMH, et al. Optimization of health care services with limited resources. *Int J Adv Sci Eng Inf Technol*. 2019;9(4):1444–9.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.